GiSMO 2025:

Factorisations
(Algebra Lecture)

by Leia Mayssa

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1 Expansion

Expansion = Multiplying out.
Expand $(a+b)(c+d)$.

2 Factorisation

Factorisation is the **reverse** of expansion.
Factorise the following expressions:

1. a^2+2a+1
2. a^2-4a+4
3. a^2-5a+6
4. $2a^2+5a-3$
5. a^2-b^2 (Difference of squares)
6. a^3-b^3 (Difference of cubes)
7. a^3+b^3 (Sum of cubes)
8. a^4-b^4
9. a^4+a^2+1
10. a^4+4b^4 (Sophie Germain)
11. $a^4-3a^2b^2+b^4$
12. $(a+b)(a-b)-4(b+1)$
13. $a^2-b^2+2(a+3b-4)$

3 Problems

Example of simple algebra problems involving factorisations:

1. Evaluate $\sqrt{111113^2-888888}$.
2. If $a^2-b^2=20$ and $a-b=4$, find the value of $a+b$.

Factorisations are also extremely useful in number theory!